



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Skill Progress Analytics with Personalized Learning Path Recommendation Engine

Course Name :Java Programming

Course Code :CSA0925

Guided by

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By

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INTRODUCTION

- ❑ **Skill Progress Analytics with Personalized Learning Path Recommendation Engine** is designed to analyze and track students' learning progress.
- ❑ The system collects **quiz scores, completed courses, and learning activities** to evaluate student performance.
- ❑ It identifies the student's **strengths and weak areas** through skill analysis.
- ❑ Based on the analysis, the system provides **personalized learning paths, course/topic recommendations, and improvement suggestions**.
- ❑ The system presents the results through a **dashboard and reports**, helping students understand their progress and plan their learning effectively



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Problem Statement

- Students often lack a **clear understanding of their learning progress and skill level** across different courses and topics.
- Existing learning systems may provide course content but do not effectively **identify individual strengths and weak areas** based on performance.
- Students may find it difficult to decide **which course or topic to learn next**, leading to an unstructured learning process.
- There is a need for a system that can **analyze quiz scores, completed courses, and learning activities** to determine student progress.
- Therefore, the proposed system aims to provide **personalized learning paths and recommendations** based on each student's progress and skill analysis.



OBJECTIVES



- ❑ **To track student learning progress** based on quiz scores, completed courses, and learning activities.
- ❑ **To analyze student skills** and identify their strengths and weak areas.
- ❑ **To calculate and evaluate overall learning performance** for each student.
- ❑ **To generate personalized learning paths** based on individual performance and skill gaps.
- ❑ **To recommend suitable courses and topics** and provide progress reports through a dashboard.

EXISTING SYSTEM VS PROPOSED SYSTEM

Existing System	Proposed System
Fixed or standard learning paths	Personalized learning paths
Limited student progress analysis	Continuous progress tracking
Difficult to identify weak areas	Identifies strengths and weak areas
Generic course recommendations	Performance-based recommendations
Limited integrated reporting	Dashboard and personalized reports



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Requirements

Requirements

Functional Requirements

- User Login
- Registration
- Course Management
- Progress Tracking
- Recommendation Engine
- Reports

Non-Functional Requirements

- Security
- Reliability
- Scalability
- Performance
- Usability

Hardware

- Intel i5/i7 Processor
- 8GB RAM
- 256GB SSD

Software

- Java
- MySQL
- JDBC
- NetBeans / Eclipse
- Windows 10/11

Dataset/APIs

Student learning records
Quiz scores
Course completion data



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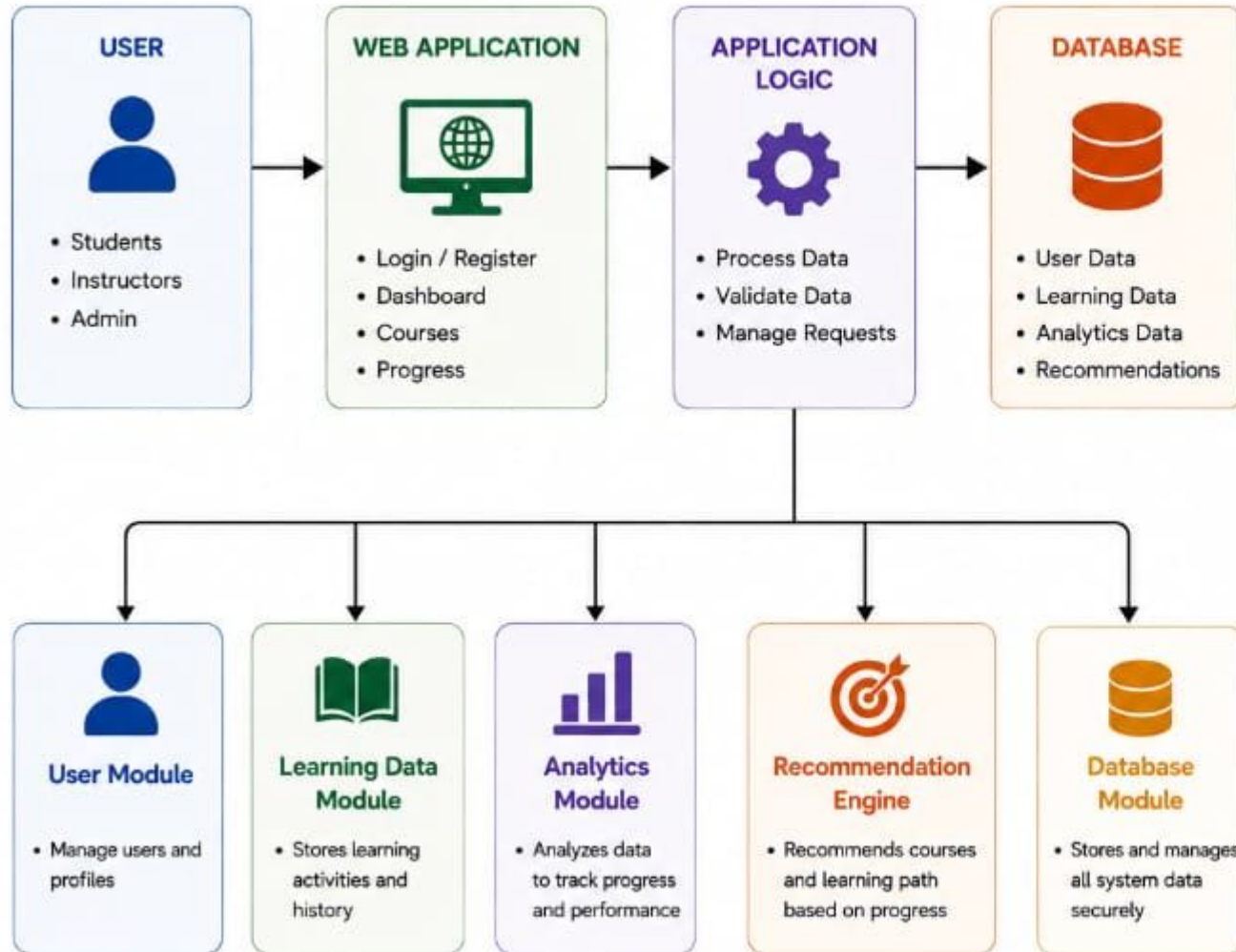
LITERATURE SURVEY

S.No	Author Names	Year	Research Problem	Tools / Techniques Used	Research Gap
1	Joseph et al.	2022	Fixed paths ignore individual learning styles.	Learning Analytics, Learning Style Model	Limited dynamic personalization
2	Raj & Renumol	2022	Learning paths need real-time adaptation.	Real-Time Learning Analytics	Limited continuous learner-state adaptation
3	Diao et al.	2022	Students need recommendations for weak concepts.	Weak Concept Mining	Limited integration with overall progress analytics
4	Zhang et al.	2023	Difficult to select suitable resources from large e-learning content.	Knowledge Graph, GCN	Limited dynamic skill-based recommendation
5	Gu	2025	Fixed paths do not suit different learner abilities.	Graph Attention, Deep RL	High complexity and data requirements



SYSTEM ARCHITECTURE

BLOCK DIAGRAM



WORKFLOW



MODULE DESIGN

1. User & Course Management Module

- ❑ **Student Registration:** Allows new students to create their accounts.
- ❑ **Login Authentication:** Validates student login credentials.
- ❑ **Course Management:** Manages available courses and course information.
- ❑ **Learning Activity Management:** Maintains students' learning activities.
- ❑ **Data Storage:** Stores student, login, course, and learning-related information.



MODULE DESIGN

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SIMATS Skill Progress Analytics
Personalized Learning Path Recommendation Engine • CSA0925 Java Programming

Active Student: P. Jeevan Kumar (192472082) ▼ [Login / Register](#)

[Student Dashboard](#) [Course & Learning Hub](#) [Progress & Skill Analytics](#) [Personalized Learning Path](#) [Student Report \(Slide 7\)](#) [UML & Database Design](#)

JAVA PROGRAMMING - CSA0925
Student Learning Dashboard
Track student progress, identify strengths and weak areas, and discover personalized learning recommendations. [Student Report](#)

3
Total Students

6
Total Subjects

69.2%
Average Score

1
Weak Subjects

Subject Performance

Subject	Score (%)
Java Programming	80%
Python	65%
Data Structures	40%
DBMS	75%
Cloud Computing	60%
Data Warehousing	75%

Overall Progress

Category	Percentage
Completed Progress	69.2%
Remaining Syllabus	30.8%

Student Management
Student registration and student information

Add Student
Student Name

Email

Password

Department

Register No

[Add Student](#)

Registered Students [Refresh List](#)

ID	NAME	EMAIL	DEPARTMENT	REPORT
1	P. Jeevan Kumar	jeevan@example.com	CSE	View Report
2	S. Dharanish	dharanish@example.com	CSE	View Report
3	Akileshwaran A	akilesh@example.com	CSE	View Report
4	praneeth	praneeth@example.com	CSE	View Report



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Module 2: Progress & Skill Analytics



- ❑ Records student quiz scores to evaluate performance.
- ❑ Tracks completed courses and learning activities.
- ❑ Calculates the overall learning progress of each student.
- ❑ Analyzes student skills based on their performance.
- ❑ Identifies the strengths of each student.
- ❑ Identifies weak areas that require improvement.
- ❑ Generates progress and skill analysis results.



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MODULE DESIGN



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Skill Progress Analytics

Personalized Learning Path Recommendation Engine • CSA0925 Java Programming

Active Student: P. Jeevan Kumar (192472082) ▼

Login / Register

Student Dashboard

Course & Learning Hub

Progress & Skill Analytics

Personalized Learning Path

Student Report (Slide 7)

UML & Database Design

Skill Analytics & Diagnostic Engine

Detailed multi-dimensional assessment of strengths, average competencies, and weak areas.

STRONG: $\geq 75\%$

AVERAGE: 60-74%

WEAK: $< 60\%$

Java Programming

STRONG (82%)

Mastery Status: Advanced Mastery

Identified Strengths:

✓ Core Concepts & Syntax ✓ Problem Solving & Design

Python

AVERAGE (68%)

Mastery Status: Intermediate Proficiency

Identified Strengths:

✓ Fundamental Concepts

Weak Topics for Revision:

⚠ Advanced Scenarios & Optimization

Data Structures

WEAK (45%)

Mastery Status: Needs Remedial Support

Identified Strengths:

Weak Topics for Revision:

⚠ Dynamic Programming ⚠ Binary Search Trees (BST)
⚠ Graph Algorithms (Dijkstra)

DBMS

STRONG (78%)

Mastery Status: Advanced Mastery

Identified Strengths:

✓ Core Concepts & Syntax ✓ Problem Solving & Design

Cloud Computing

AVERAGE (62%)

Mastery Status: Intermediate Proficiency

Identified Strengths:

✓ Fundamental Concepts

Weak Topics for Revision:

⚠ Advanced Scenarios & Optimization

Data Warehousing

STRONG (80%)

Mastery Status: Advanced Mastery

Identified Strengths:

✓ Core Concepts & Syntax ✓ Problem Solving & Design

Progress & Skill Analytics Process Flow (Slide 5)

1. Input:

Quiz scores, completed courses, and learning activity logs.

2. Process:

Analyzes student performance and calculates weighted learning progress.

3. Output:

Progress status, strengths, and specific weak topic identification.

3. Recommendation & Dashboard Module

- Generates personalized learning paths based on student performance.
- Recommends suitable courses and topics for each student.
- Provides suggestions to improve identified weak areas.
- Displays student progress through a dashboard.
- Generates personalized progress and skill reports.



MODULE DESIGN

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SIMATS Skill Progress Analytics
Personalized Learning Path Recommendation Engine • CSA0925 Java Programming

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Student Dashboard Course & Learning Hub Progress & Skill Analytics Personalized Learning Path Student Report (Slide 7) UML & Database Design

Personalized Learning Path Recommendation Engine

Adaptive learning paths and prioritized action items customized to student weak areas.

Enrichment

Industry Readiness: Java Programming
Subject: Java Programming • Topic: Capstone & Project Implementation
Apply concepts in a real-world mini-project to build comprehensive portfolio-grade engineering artifacts.

Recommended Learning Resources:

- SIMATS Industry Case Studies
- Open Source Project Contribution Guide

~5 Hours Required

Medium Priority

Bridge to Excellence: Python
Subject: Python • Topic: Advanced Applications & Problem Sets
Review optimization techniques and attempt medium-difficulty problem sets to elevate mastery into Strong (>75%).

Recommended Learning Resources:

- Advanced Study Guide: Python
- Timed Mock Assessment & Feedback

~6 Hours Required

High Priority

Mastery Acceleration: Data Structures
Subject: Data Structures • Topic: Dynamic Programming
Step 1: Complete visual conceptual review. Step 2: Solve 10 guided coding practice exercises. Step 3: Retake topic assessment to reach >75%.

Recommended Learning Resources:

- <https://simats.edu/library/data-structures-remedial>
- Interactive Algorithm Visualizer & Sandbox
- Module Video Lectures - Part 1 & 2

~12 Hours Required

High Priority

Intensive Problem Solving: Binary Search Trees (BST)
Subject: Data Structures • Topic: Binary Search Trees (BST)
Focus on step-by-step problem decomposition and debug sample test cases before moving to complex problems.

Recommended Learning Resources:

- Hands-on Lab Worksheet - Unit Data Structures
- Practice Quiz & Solution Walkthrough

~8 Hours Required

Enrichment

Industry Readiness: DBMS
Subject: DBMS • Topic: Capstone & Project Implementation
Apply concepts in a real-world mini-project to build comprehensive portfolio-grade engineering artifacts.

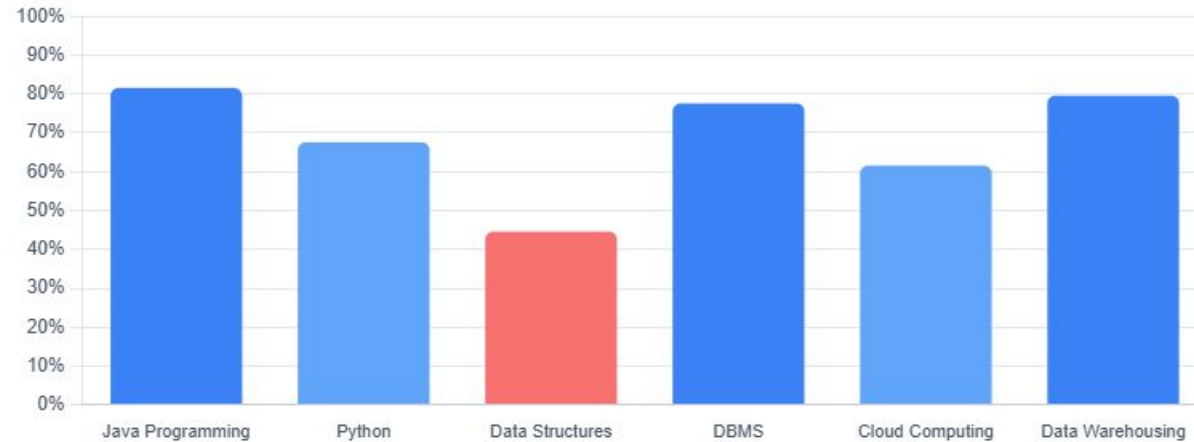
Recommended Learning Resources:

~5 Hours Required

RESULT AND PERFORMANCE ANALYSIS

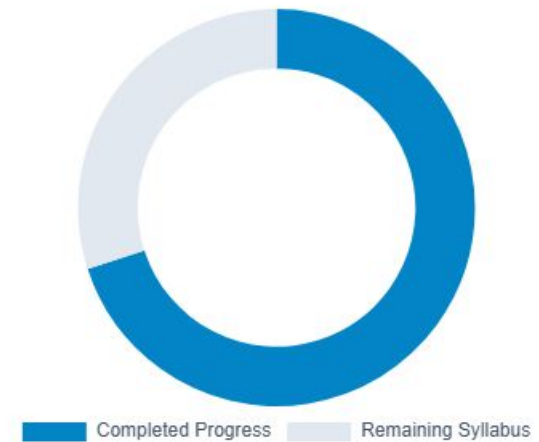
Subject Performance

QUIZ & MODULE SCORES



Overall Progress

SYLLABUS COMPLETION



PROJECT HOSTING AND DEPLOYMENT

- ❑ The project source code is maintained and version-controlled using **GitHub**.
- ❑ The backend application is connected to **GitHub** for deployment and source-code management.
- ❑ The backend is hosted on **Render**, providing a publicly accessible web service.
- ❑ GitHub changes can be integrated with Render to **build and deploy updated versions** of the backend.
- ❑ This setup provides a simple and reliable process for **code management, backend hosting, and deployment**.



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CONCLUSION

- ❑ The system successfully tracks and analyzes student learning progress.
- ❑ It identifies student strengths and weak areas through performance analysis.
- ❑ Personalized learning paths and suitable course recommendations are generated.
- ❑ The dashboard provides clear progress, skill analysis, and recommendation reports.
- ❑ The system helps students follow a more **structured and personalized learning process**.

REFERENCES

- Joseph, L., Abraham, S., Mani, B. P., & Rajesh, N. (2022). *Learning Analytics Intervention for Personalized Learning*.
- Raj, N. S., & Renumol, V. G. (2022). *Real-Time Learning Analytics for Adaptive Learning Path Recommendation*.
- Diao, et al. (2022). *Personalized Learning Path Recommendation Based on Weak Concept Mining*.
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- Gu, R. (2025). *Personalized Learning Path Recommendation Using Graph Attention and Deep Reinforcement Learning*.
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HOST LINK

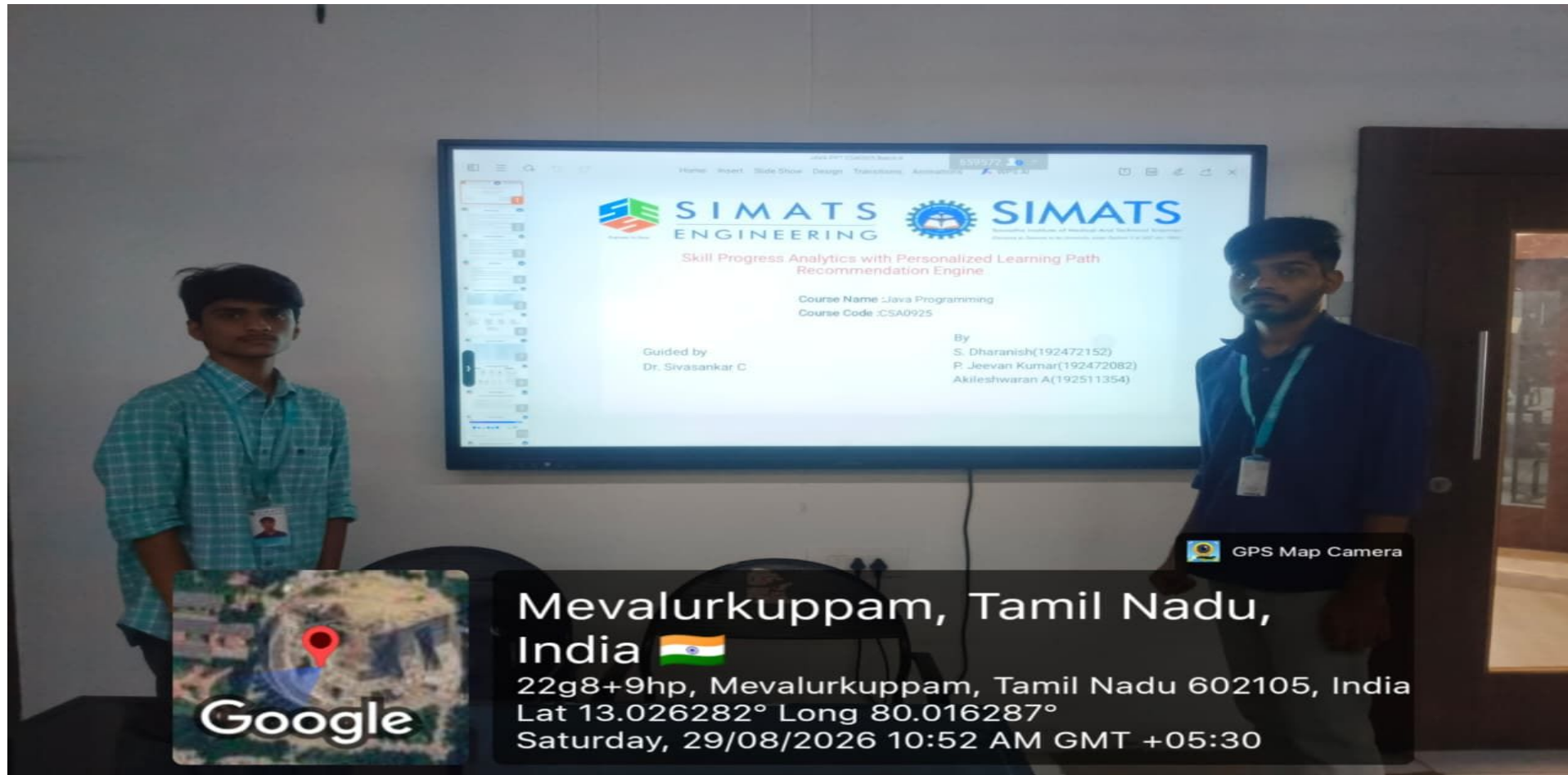


<https://java-capstone.onrender.com>



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GEO TAG



THANK YOU